

The Danger of Blind Data: An Exploratory Data Analysis on Student Burnout

Abstract

This project examines a 150,000-row student burnout dataset from Kaggle using Python, Pandas, and Seaborn. The initial objective was to build a machine learning model predicting burnout based on screen time and sleep. However, Exploratory Data Analysis (EDA) revealed uniform data distributions and a near-zero correlation matrix ($r \approx 0$). This confirmed the dataset was synthetic, shifting the project focus to data validation and integrity.

Introduction

Since Computer Science is one of my interests, I wished to create a data analysis project to add to my portfolio while learning more about data analysis. During my search for project topics, I discovered a Kaggle dataset on student mental health and burnout. As burnout is a common phenomenon among students, I thought that it would be interesting to look into it.

At first, my intention was to explore correlations between different aspects of life such as sleep duration, screen time, number of studying hours, and academic pressure, and student burnout. Moreover, I wished to acquire skills in data visualization, analysis, and cleaning using such programs as Python, Pandas, Matplotlib, and Seaborn.

However, as I proceeded with the work, I realized that some results I got seemed too perfect and unrealistic. Instead of using the data as it is, I started to wonder whether the dataset itself could be considered reliable. The change in the focus of the project allowed me to research not only student burnout but also the integrity of data – what data analysts do to find out whether the dataset is reliable enough.

What started as a portfolio project turned out to be an important lesson on the need for data validation. One needs to be sure that the data is valid before conducting an analysis or interpretation of it.

2. Methodology & Stack

Methodology

The analysis was done in Google Colab, an online cloud environment that uses Jupyter Notebook for Python. I downloaded the dataset from Kaggle in the form of a .zip file. It was uploaded and decompressed into the notebook to get the .csv file. Pandas was used for loading, cleaning, examining, and analyzing the dataset. Visualization of the data was done using libraries such as Matplotlib and Seaborn to produce scatter plots and heatmaps.

3. Data Analysis & Findings

Finding 1: Overplotting and Uniform Distribution

Firstly, I created a scatter plot between `screen_time_hours` and `daily_sleep_hours`. Since there were around 150,000 data points in the dataset, many points overlapped, resulting in severe overplotting and making the plot look like a solid block without any pattern at all.

In order to solve this problem, I plotted a 2D histogram heatmap using the Seaborn `histplot()` function. I expected to find some clusters of the data points for the common sleep time since, in real-world data, we usually have some areas that have a high density of the observation points. However, there was a near-uniform distribution throughout the whole plot and there were no visible clusters.

Finding 2: The Correlation Matrix

To further investigate the dataset, I selected the numerical variables and calculated a Pearson correlation matrix, visualizing the results with Seaborn's `heatmap()`.

The Pearson correlation coefficient measures the strength and direction of the linear relationship between two variables:

$$r = \frac{\sum (x_i - \bar{x}) (y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

As expected, every feature had a perfect correlation with itself along the diagonal ($r = 1.0$), however, the correlation coefficients of nearly every other pair of features were near to zero. While this finding does not prove anything alone, the lack of any correlations between those features which would be likely to be related indicated that this data set was probably artificially created, not gathered from actual students.

4. Conclusion & Lessons Learned

My initial intention was to conduct a thorough data analysis in order to have a strong analytical paper in my portfolio and to develop my skills in Python programming and Data Science in general. In the end, my project took a completely different route from the one that I had initially intended.

When examining the data, I began noticing certain anomalies that seemed unrealistic. Instead of conducting further analysis, I decided to verify whether the dataset I was working with could be considered reliable. As a result, this step became the most important part of my project.

Beyond learning how to apply libraries such as Pandas, Seaborn, or Matplotlib, I realized that doing real Data Science involves questioning the very data itself. Regardless of how elaborate a model might be, its reliability depends entirely on the data that it is based on.

Even though the project did not end up where I had initially intended it to go, through this project, I have gained a far more valuable insight into data integrity and the importance of critical thinking. Reflecting upon the project, I can say that I have learned how to be a better programmer, better

problem solver, and even someone who questions the outcomes before accepting them. This is what I will learn from for my future Computer Science projects.